

The background of the entire slide is a dense, repeating pattern of Kirby's face. Each Kirby face is pink with large, expressive blue eyes and a small, open mouth. The faces are arranged in a way that creates a sense of depth and movement, with some faces appearing more prominent than others.

Kir-B

Final Presentation

Ola, Elysia, Lindsay, Evan, Miles, Nathan

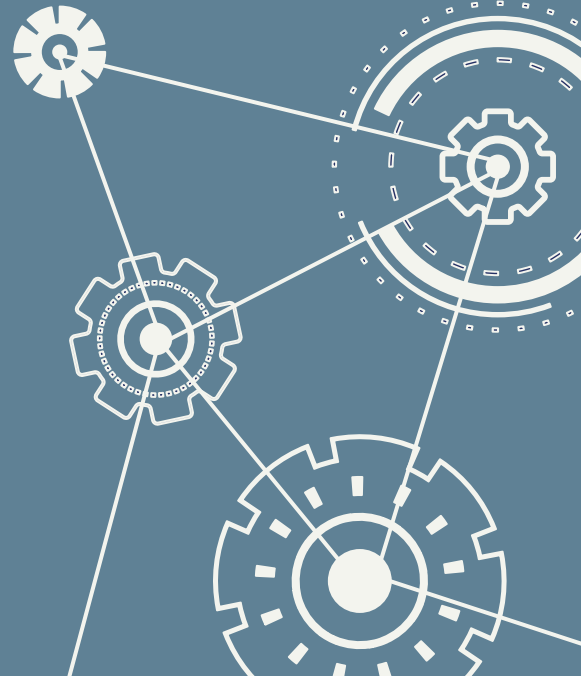
Criteria

Presentation Structure

- be ready to present CAD and manufacturing results (quality of your manufactured parts) for all parts of the yo-yo
 - explain your design theming
 - review the results of your manufacturing processes and highlight the quality of your final yo-yos
 - bring example yo-yos to show off your work!
- students will be graded on
 - completeness of the required elements
 - clarity of presentation

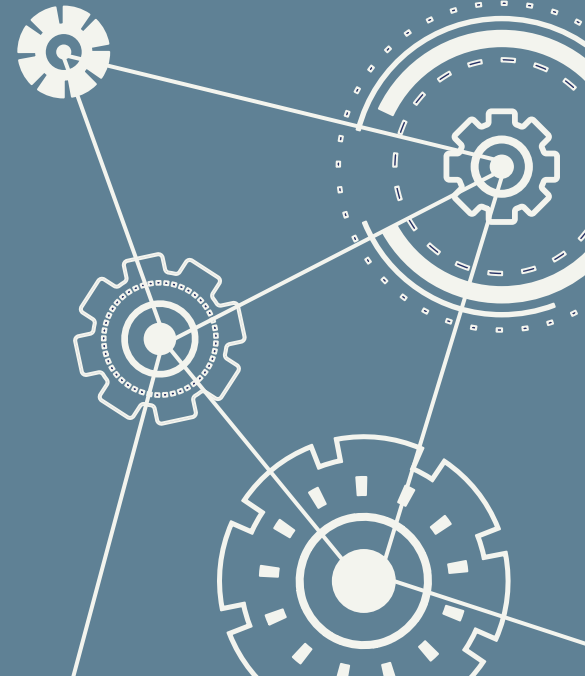


Design Theming





Result of Manufacturing Processes





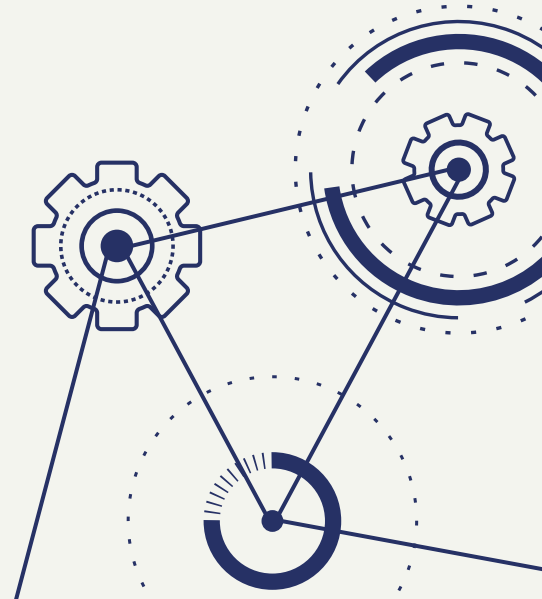
Process 1: IM Floatie



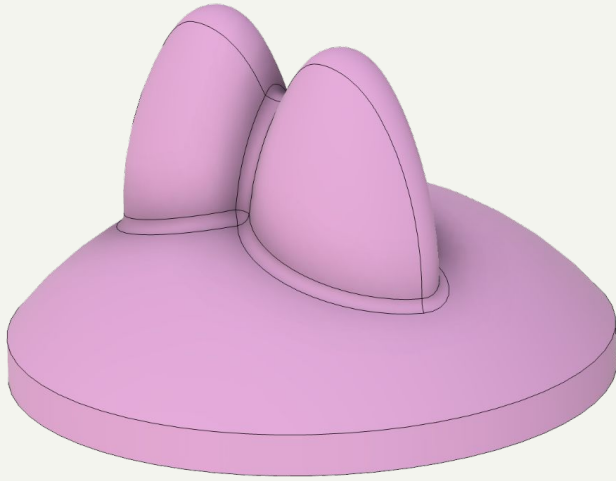
CAD press fit diameter: 2.5 in



Avg measured press fit diameter: 2.505 in



Process 2: IM Lower Body



CAD press fit: 2.524 in



Avg measured press fit: 2.512 in



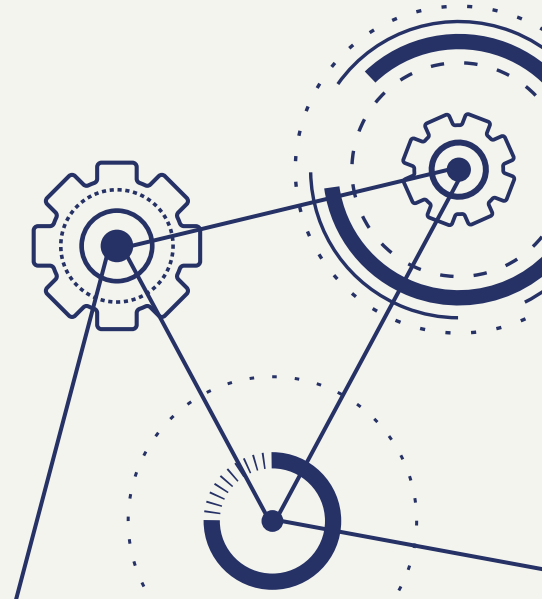
Process 3: IM Upper Body



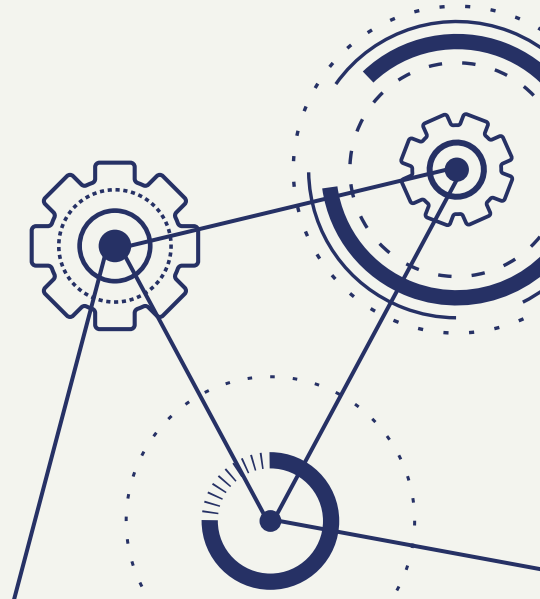
CAD press fit: 2.524 in



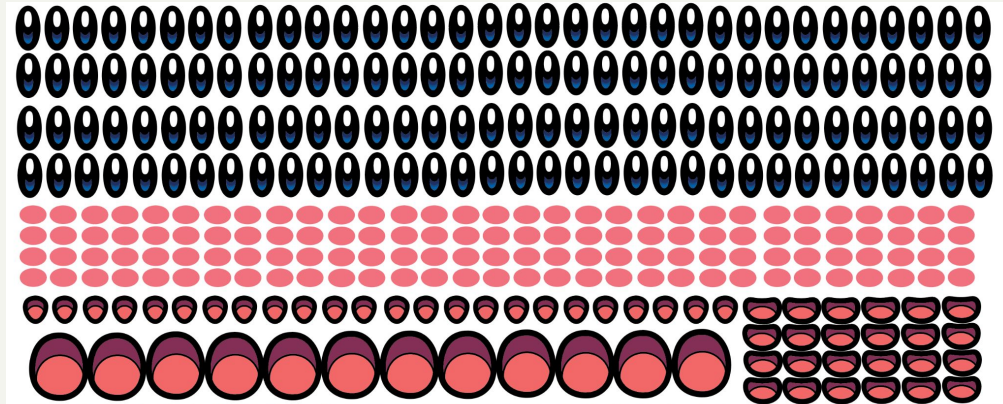
Avg measured press fit: 2.508 in



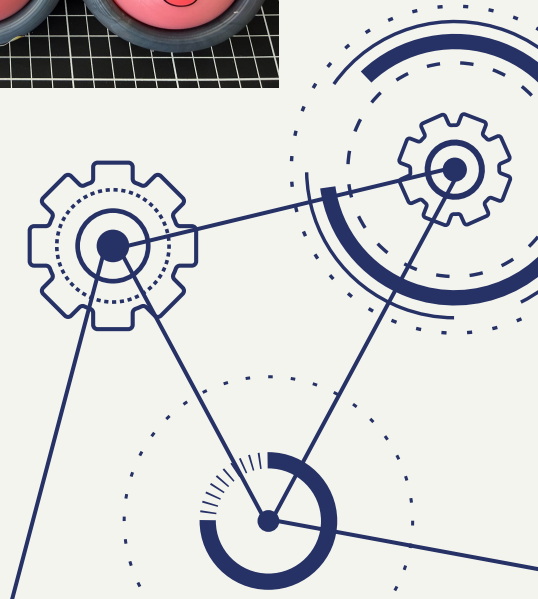
Process 4: Thermoform



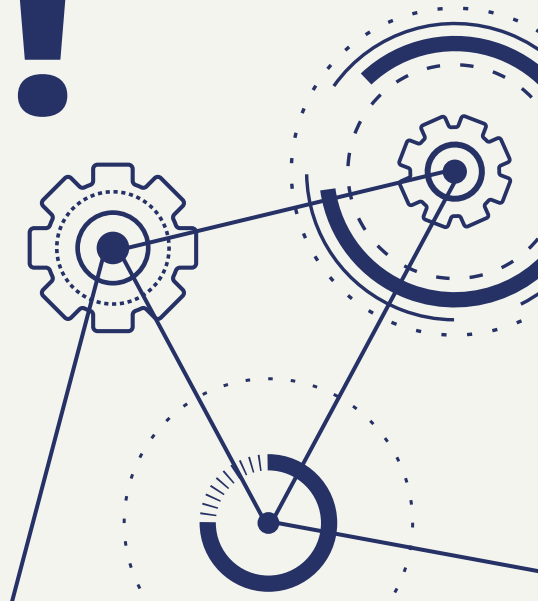
Process 5: Stickers!



sticker sheet :p



IT YO'S!!!



Criteria

Press Fit Engineering Analysis

- present the analysis documented in the Manufacturing Spreadsheet, preparing slides that highlight each key decision making step
- explain the full thread of the press-fit features of your yo-yo, connecting the decisions made during the design phase to the results of the manufacturing phase
 - how well did you deliver on your own specifications?
 - show your **specification limits** on the histogram of your measured yo-yo parts to compare the variation in your process to the tolerances you specified (**process capability**)
 - how do your body and ring distributions characterize the success of your manufacture processes?
 - using engineering reasoning, why are there differences between your specifications and measured parts?
 - what does your **interference distribution** look like?
 - what did you learn along the way?
- students will be graded on:
 - the completeness of the spreadsheet parameters up to this point
 - the soundness of the engineering reasoning used to make key decisions about press fit parameters
 - consistency between the manufacturing spreadsheet, CAD, and the message conveyed in the presentation



Press Fit & Shrinkage Analysis



Design Decisions → Manufacturing Result

● Design decision	● Manufacturing Result
● Use the same interference for the upper and lower body	● Different geometries → different shrinkage → different interferences with the floatie

Distributions

Dimension	Mean (in.)	Standard Deviation (in.)
OD upper body	2.508	0.008
OD lower body	2.512	0.009
ID of floatie	2.505	0.009
Interference (Upper)	0.003	0.012
Interference (Lower)	0.007	0.012

Distributions - Floatie

$$\mu = 2.505''$$

$$\sigma = 0.009''$$

$$n = 100$$

CAD SPEC: 2.500" +/- 0.050"

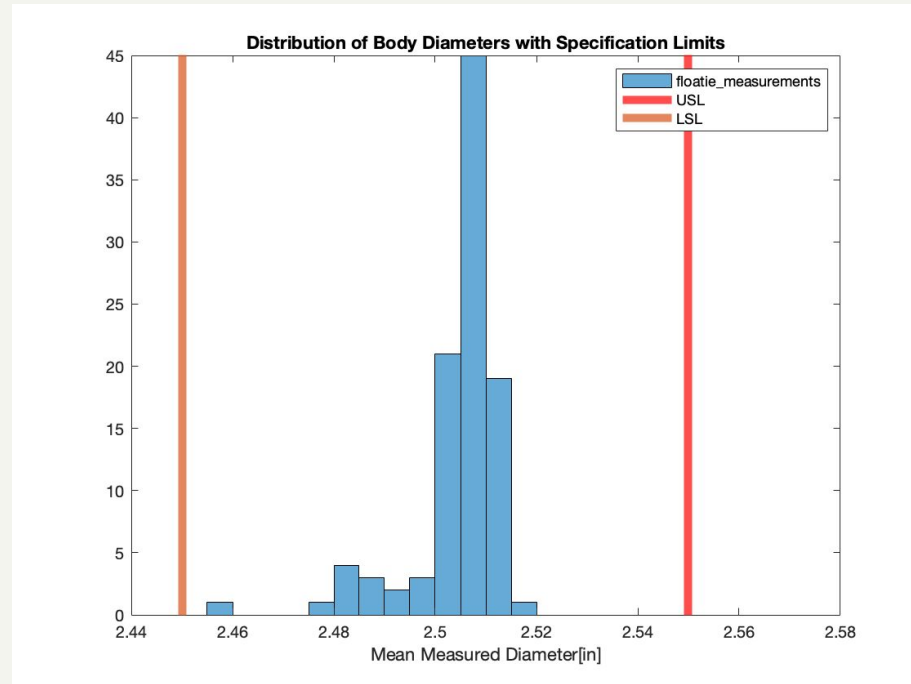
USL = 2.550"

LSL = 2.450"

Process Capability:

$$C_p = 1.841$$

$$C_{pk} = 1.672$$



Distributions - Lower body

$$\mu = 2.512''$$

$$\sigma = 0.009''$$

$$n = 50$$

CAD SPEC: 2.524" +/- 0.050"

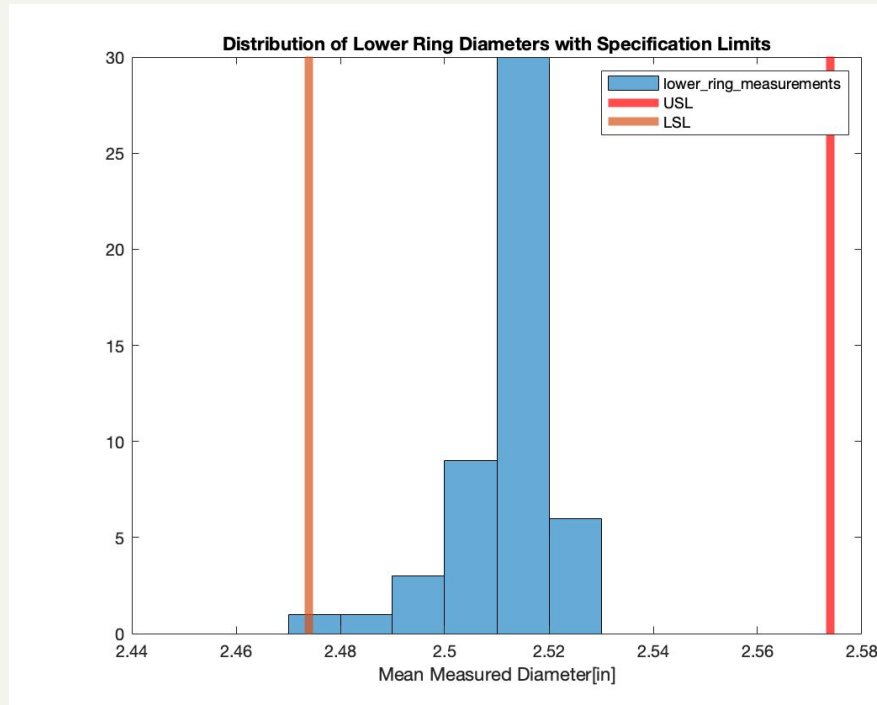
USL = 2.574"

LSL = 2.474"

Process Capability:

$$C_p = 1.938$$

$$C_{pk} = 1.468$$



Distributions - Upper body

$$\mu = 2.508''$$

$$\sigma = 0.008''$$

$$n = 50$$

CAD SPEC: 2.524" +/- 0.050"

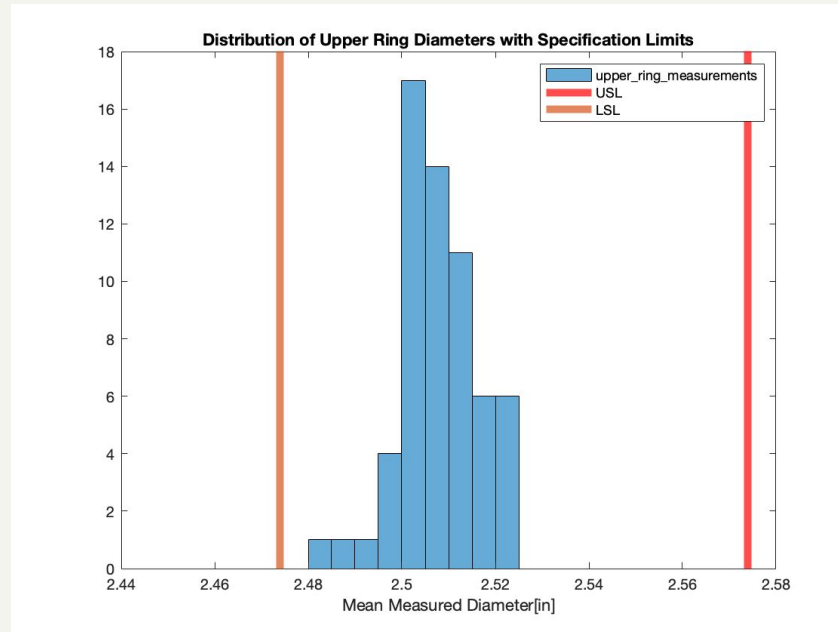
USL = 2.574"

LSL = 2.474"

Process Capability:

$$C_p = 2.020$$

$$C_{pk} = 1.352$$



Distributions - Upper body

$\mu = 2.508''$

$\sigma = 0.008''$

$n = 50$

CAD SPEC: 2.524" +/- 0.050"

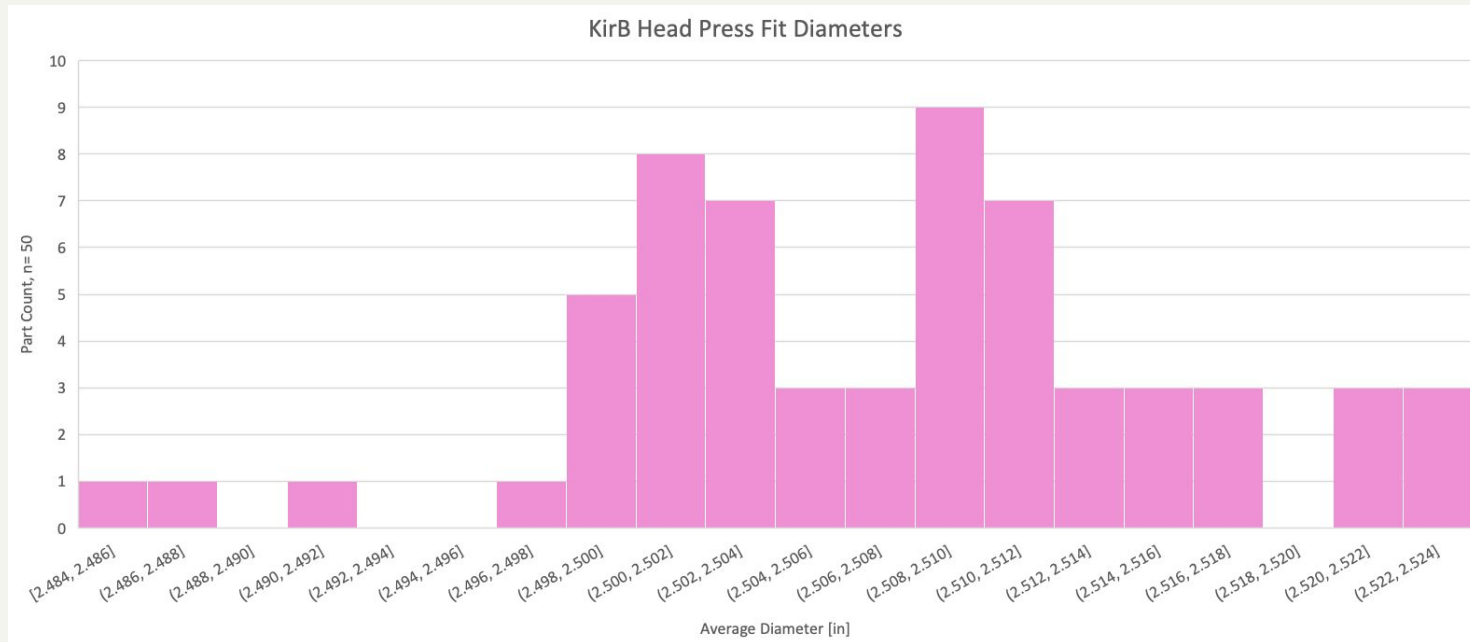
USL = 2.574" (off graph, to right)

LSL = 2.474" (off graph, to left)

Process Capability:

$C_p = 2.020$

$C_{pk} = 1.352$



Distributions - Interference

Combining Normally Distributed Data Sets

added data sets:

$$\mu_{new} = \mu_1 + \mu_2$$

$$\sigma_{new} = \sqrt{\sigma_1^2 + \sigma_2^2}$$

subtracted data sets:

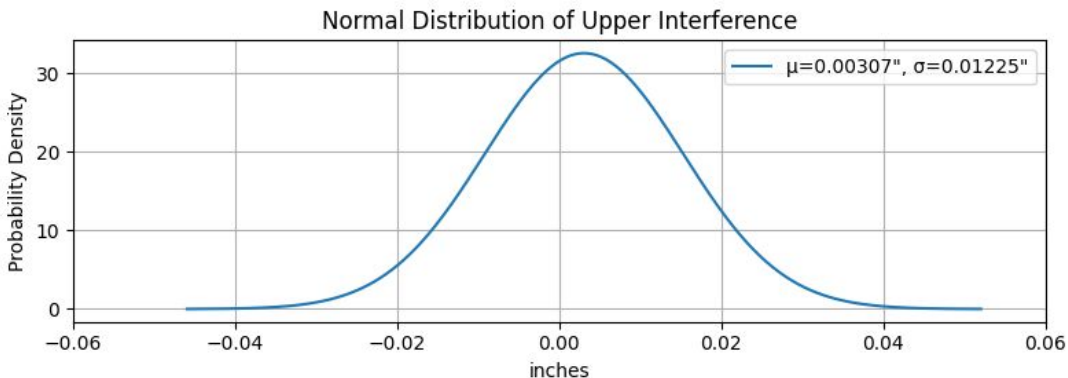
$$\mu_{new} = \mu_1 - \mu_2$$

$$\sigma_{new} = \sqrt{\sigma_1^2 + \sigma_2^2}$$

Upper Interference

$$\mu = 0.003''$$

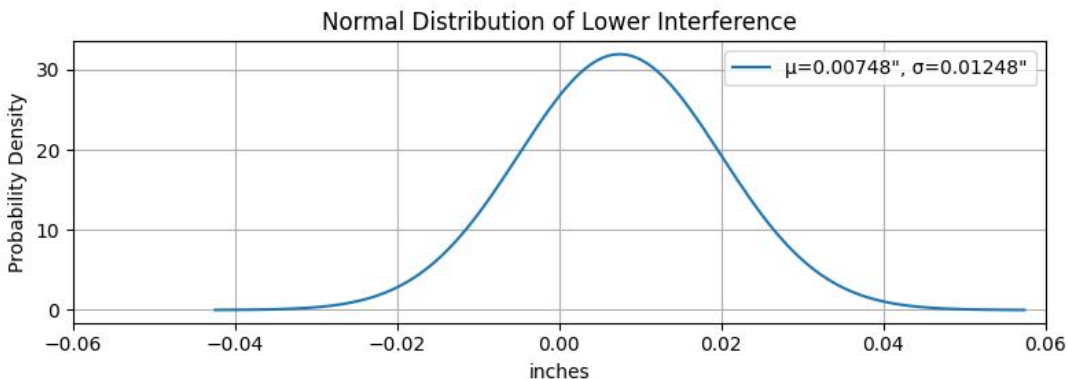
$$\sigma = 0.012''$$



Lower Interference

$$\mu = 0.007''$$

$$\sigma = 0.012''$$

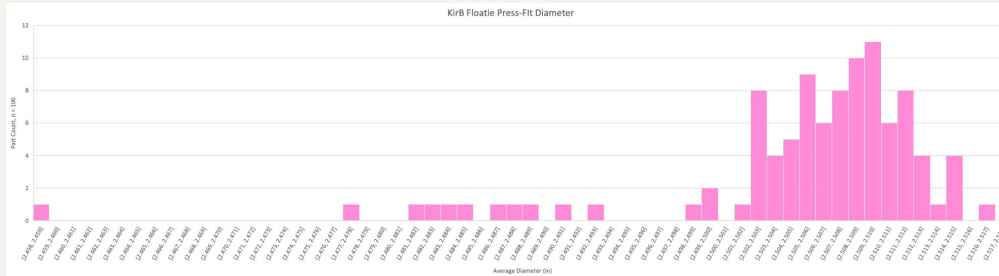


Press Fit Engineering Analysis

1. **Nominal CAD** - MATLAB calculations, past documentation
2. **Adjusted CAD** - Polyurethane specs and data sheets
3. **Machined Mold** - Ensuring upper and lower ring have the same critical dimension
4. **Injection Molding** - Determining experimental parameters

1. Nominal CAD			2. Adjusted CAD			3. Machined Mold			Injection Molding		
body press fit diameter:	2.5000	[in]	body press fit diameter:	2.538	[in]	body press fit diameter:	2.5357	[in]	body press fit diameter (avg):	2.505	[in]
ring press fit diameter:	2.5242	[in]	ring press fit diameter:	2.563	[in]	top ring press fit diameter:	2.5513	[in]	top ring press fit diameter (avg):	2.508	[in]
						bottom ring press fit diameter:	2.563	[in]	bottom ring press fit diameter (avg):	2.512073333	[in]
diameter interference:	0.0242	[in]	diameter interference:	0.025	[in]	diameter interference:	0.0156	[in]	diameter interference:	N/A	[in]
length of engagment:	0.2000	[in]	body shrinkage %:	1.500%		body deviation:	-0.0024	[in]	body std dev:	0.009051	[in]
part tolerance, upper:	0.0500	[in]	ring shrinkage %:	1.500%		upper ring deviation:	-0.0113	[in]	top ring std dev:	0.008252	[in]
part tolerance, lower:	-0.0500	[in]				lower ring deviation:	0.000	[in]	bottom ring std dev:	0.008599	[in]
									measured shrinkage, body:	1.226617239	%
									measured shrinkage, top ring:	1.71043199	%
									measured shrinkage, bottom ring:	1.986994408	%

Press Fit Lessons Learned

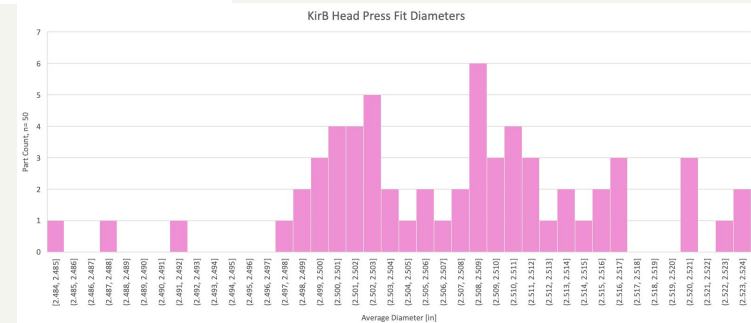
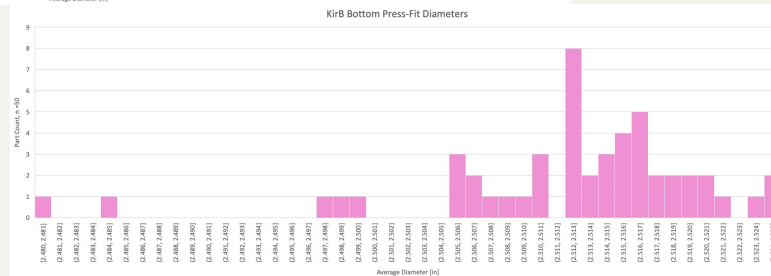


Variability introduced through

- Different people gathering measurements
- Taking measurements while pieces warm

Engineering Reasoning

- Machining tolerances/discrepancies
- Non uniform shrinkage
- Inaccurate measuring techniques



Success!

- The pieces fit together!
- 9/10 passed the drop test

Criteria

Lessons Learned

- with the rest of your presentation time, highlight the most important lessons learned during your 2.008 yo-yo project
- what were the key strategic decisions your team had to make to end up with a successful yo-yo?
- what were the most challenging parts of the project?
 - how did they match up to what you thought were going to be the hardest parts of the project?
- what advice do you have for future yo-yo teams?!?

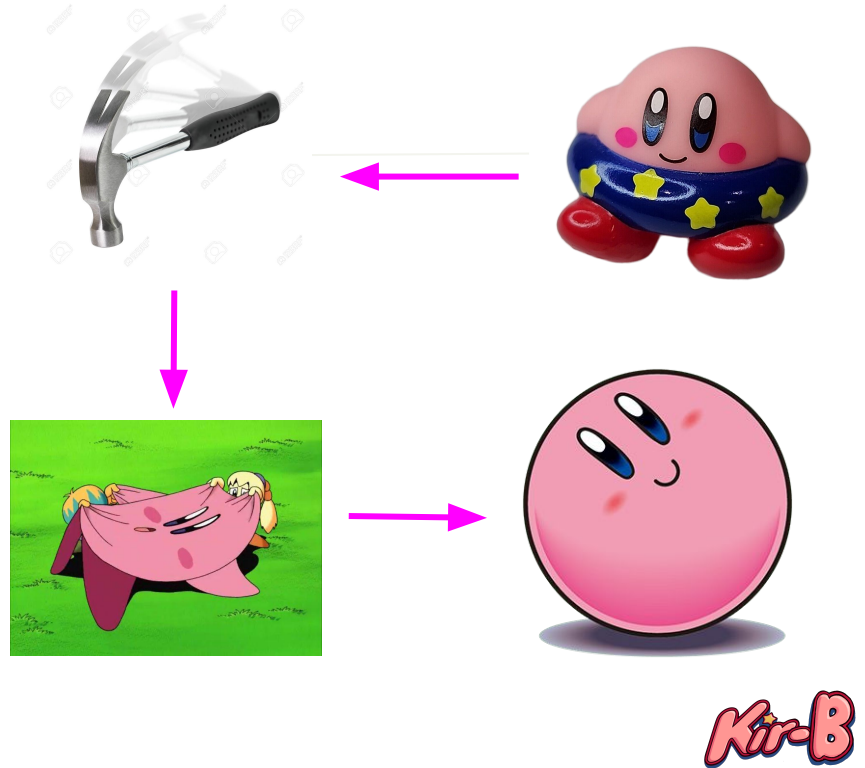
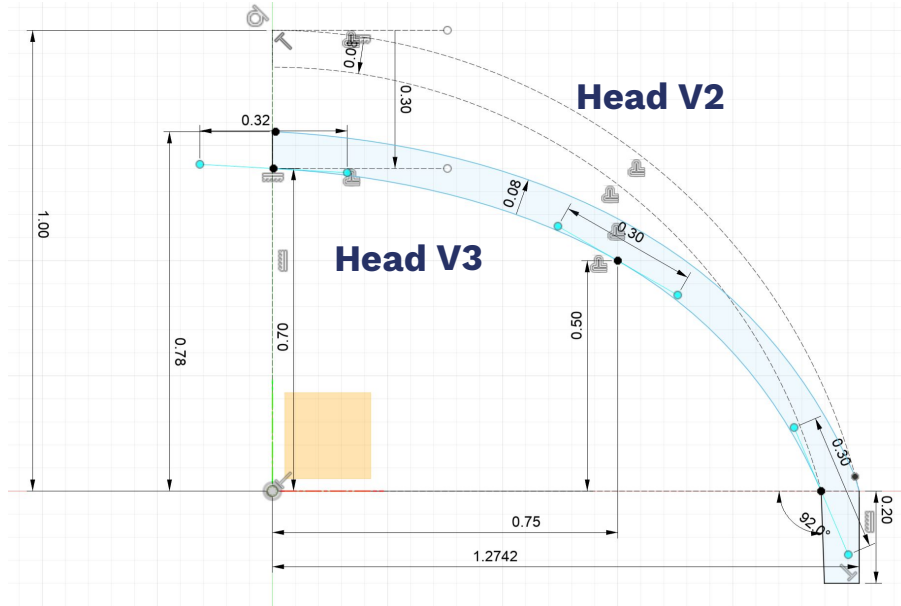


Lessons Learned & Advice



Lesson 1: Use your contingency plan

- Choose simplicity when necessary to keep progressing



Lesson 2: Be there for your team!

Team Goals:



Have all IM Parts Fabricated (4/10) **✗** - *did not happen*
Finalize all ring and body molds/solve press fit (4/17) **✓**
Assemble one complete Yo-Yo (4/25)

Lesson 3: Time-value of Decisions

Quick and
temporary fixes

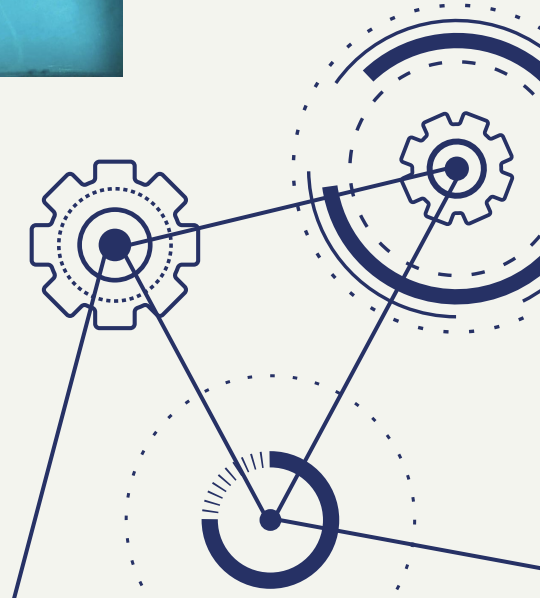
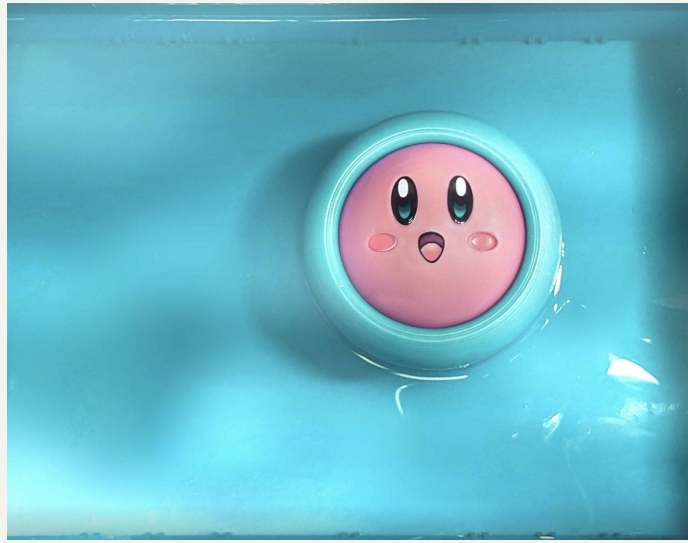
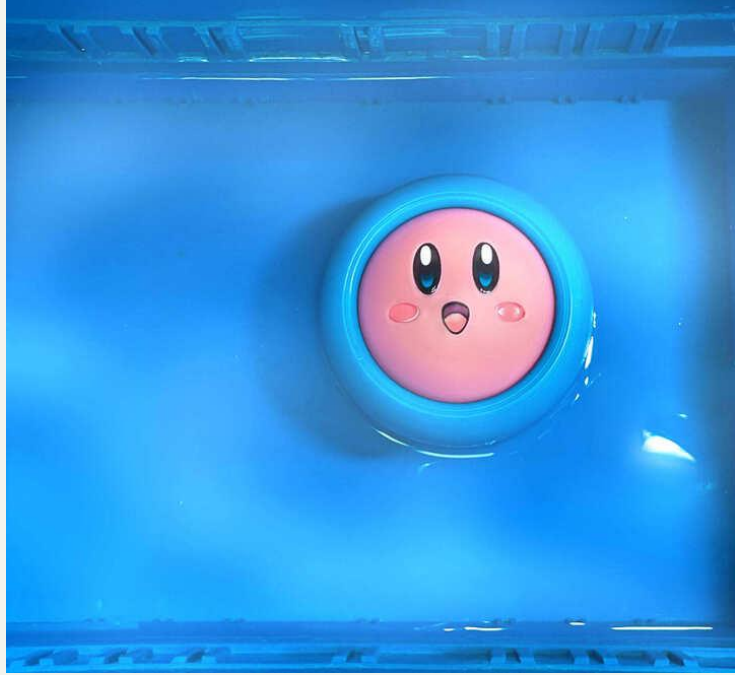
vs.

High upfront
investment, long
lasting solutions



Q & A





What's left to do?

- Eliminate ejector pin marks on kirby face
- Find the correct pink for kirby halves
- TF 50 Kirby features
- Add star stickers to floatie
- Add eye stickers to face
- Assembly

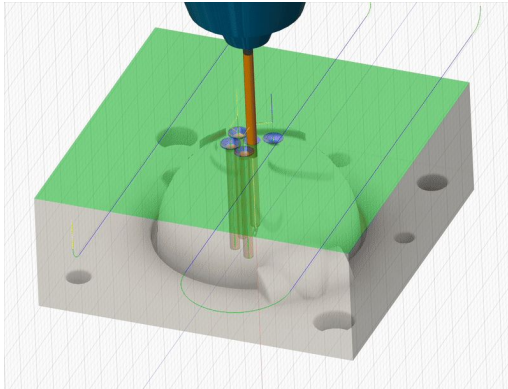


CAM Programs

Upper Body - core

Time: 2:57 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face2	1/1	Setup3	2	#14 - Ø2" 45° face
 Drill2	1/1	Setup3	2	#11 - Ø1/4" 130° spot drill (SPOT DRILL)
 Drill3	1/1	Setup3	2	#12 - Ø0.12" drill (Ejector Pin Drill)
 Adaptive2	1/1	Setup3	2	#5 - Ø1/4" flat (End Mill)
 Contour3	1/1	Setup3	2	#5 - Ø1/4" flat (End Mill)
 Morphed ...	1/1	Setup3	2	#3 - Ø1/8" flat (End Mill)
 2D Contour2	1/1	Setup3	2	#10 - Ø1/16" 2° tapered (2 Degree Tapered End Mill)
 Flat1	1/1	Setup3	2	#5 - Ø1/4" flat (End Mill)



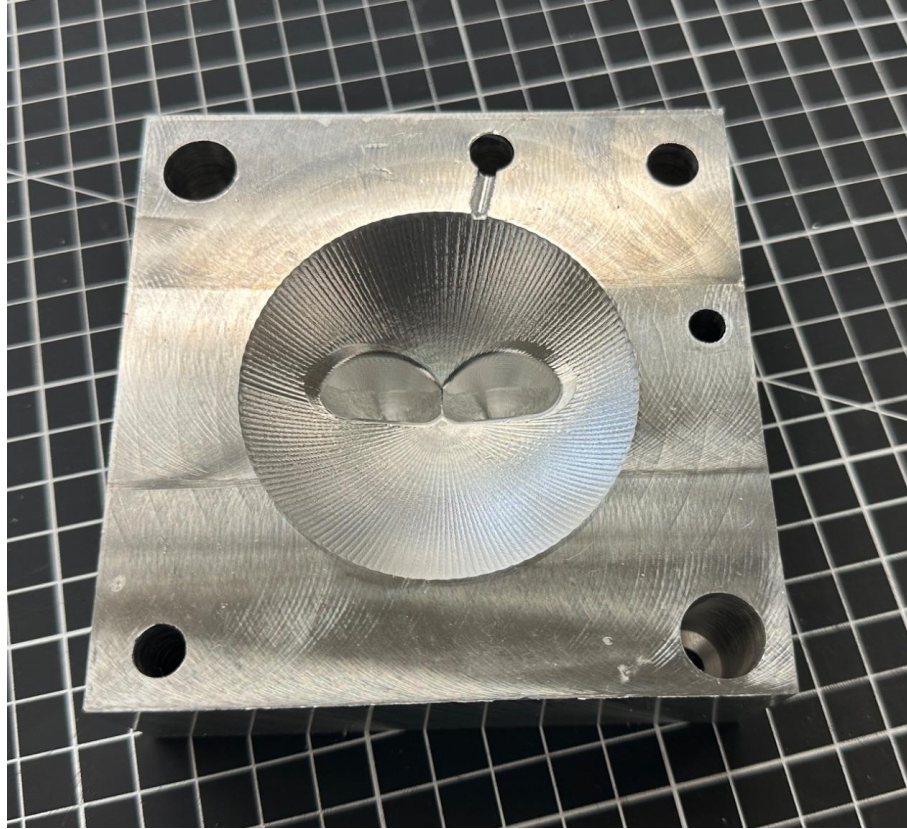
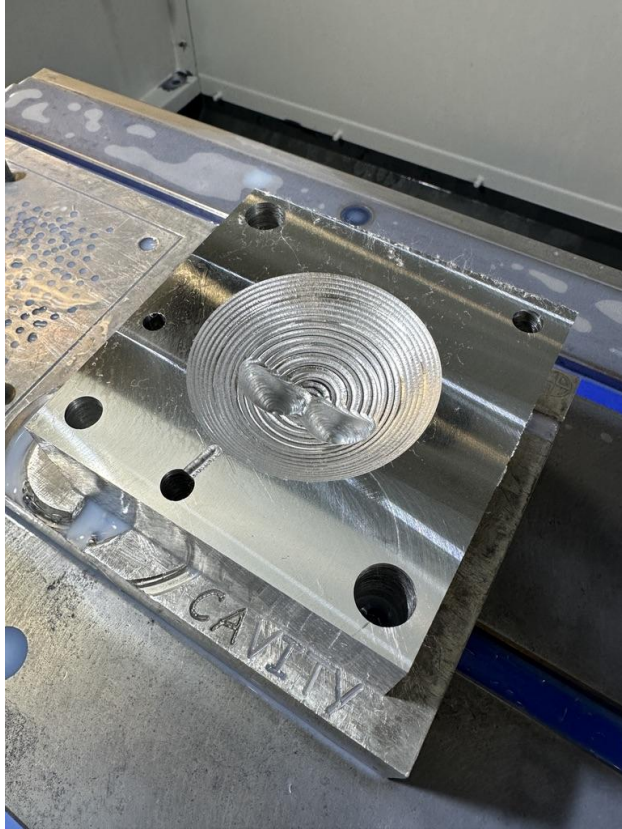
Kir-B

Floatie Difficulties

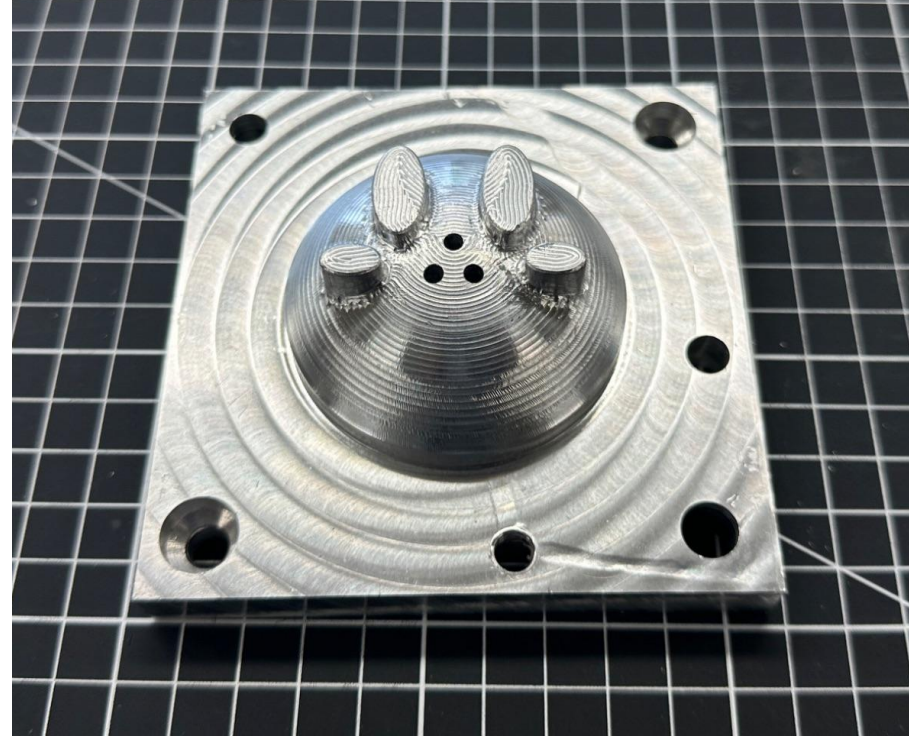
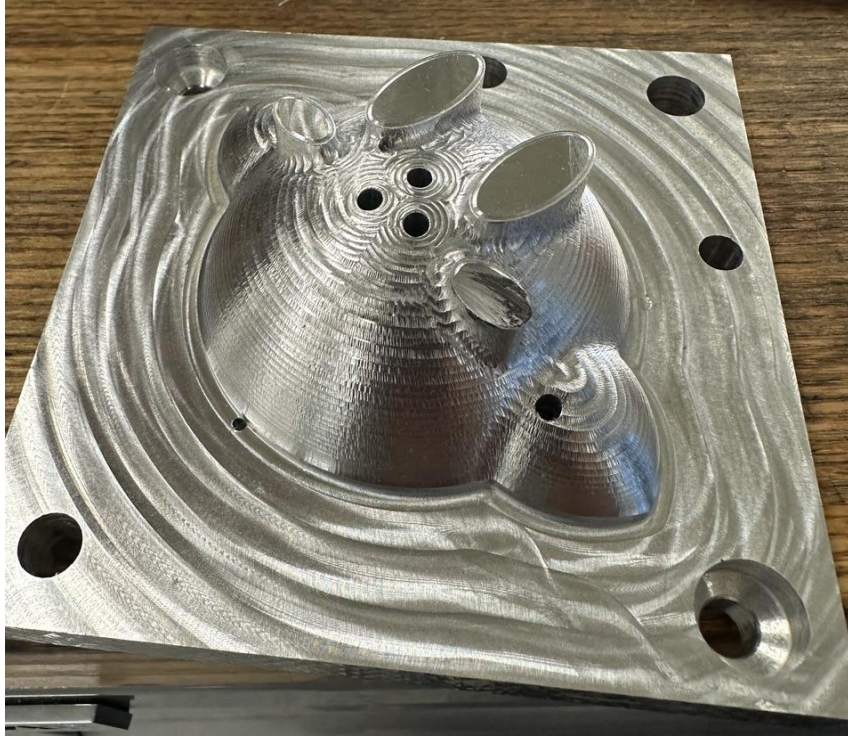


Thinnest part of mold failed under heat and pressure of plastic

Lower Body Difficulties



Upper Body Difficulties



Upper Body Difficulties



CAM Programs

Upper Body - cavity

Time: 1:35 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face1	1/1	cavity	3	#14 - Ø2" 45° face
 Adaptive1	1/1	cavity	3	#6 - Ø3/8" flat (End Mill)
 Contour1	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)
 Contour2	1/1	cavity	3	#5 - Ø1/4" flat (End Mill)
 2D Contour1	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)



CAM Programs

Lower Body - core

Time: 1:02 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face1	1/1	core	2	#14 - Ø2" 45° face
 Drill1	1/1	core	2	#11 - Ø1/4" 130° spot drill (SPOT DRILL)
 Drill2	1/1	core	2	#12 - Ø0.12" drill (Ejector Pin Drill)
 Adaptive2	1/1	core	2	#6 - Ø3/8" flat (End Mill)
 Contour2	1/1	core	2	#9 - Ø1/4" ball (End Mill)
 2D Pocket1	1/1	core	2	#10 - Ø1/16" 2° tapered (2 Degree Tapered End Mill)
 2D Contour1	1/1	core	2	#8 - Ø1/8" ball (End Mill)

CAM Programs

Lower Body - cavity

Time: 51 mins

Operation	Instance	Setup	Work Offset	Tool
 Face2	1/1	cavity	3	#14 - Ø2" 45° face
 Adaptive4	1/1	cavity	3	#9 - Ø1/4" ball (End Mill)
 Contour3	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)
 2D Contour2	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)

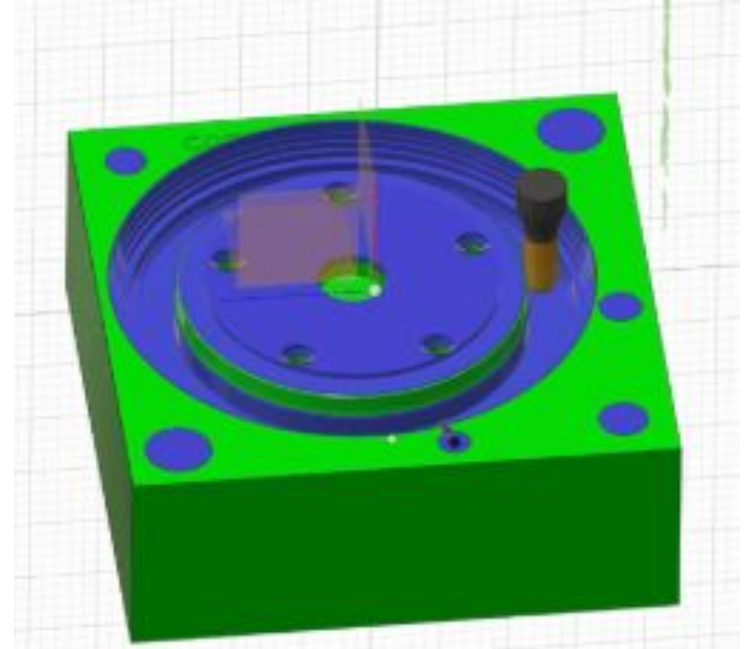


CAM Programs

Floatie - core

Time: 1:08 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face5	1/1	Setup7	2	#14 - Ø2" 45° face
 Bore1	1/1	Setup7	2	#6 - Ø3/8" flat (End Mill)
 Drill1	1/1	Setup7	2	#11 - Ø1/4" 130° spot drill (SPOT DRILL)
 Drill3	1/1	Setup7	2	#12 - Ø0.12" drill (Ejector Pin Drill)
 Adaptive7	1/1	Setup7	2	#5 - Ø1/4" flat (End Mill)
 Adaptive2	1/1	Setup7	2	#4 - Ø3/16" flat (End Mill)
 Contour4	1/1	Setup7	2	#9 - Ø1/4" ball (End Mill)
 Contour6	1/1	Setup7	2	#8 - Ø1/8" ball (End Mill)
 Flat1	1/1	Setup7	2	#6 - Ø3/8" flat (End Mill)
 2D Contour4	1/1	Setup7	2	#8 - Ø1/8" ball (End Mill)

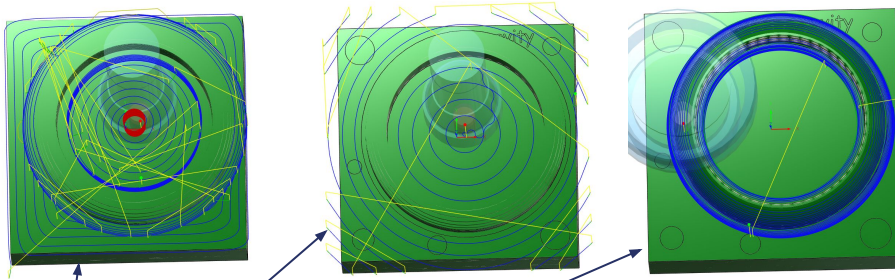


Kir-B

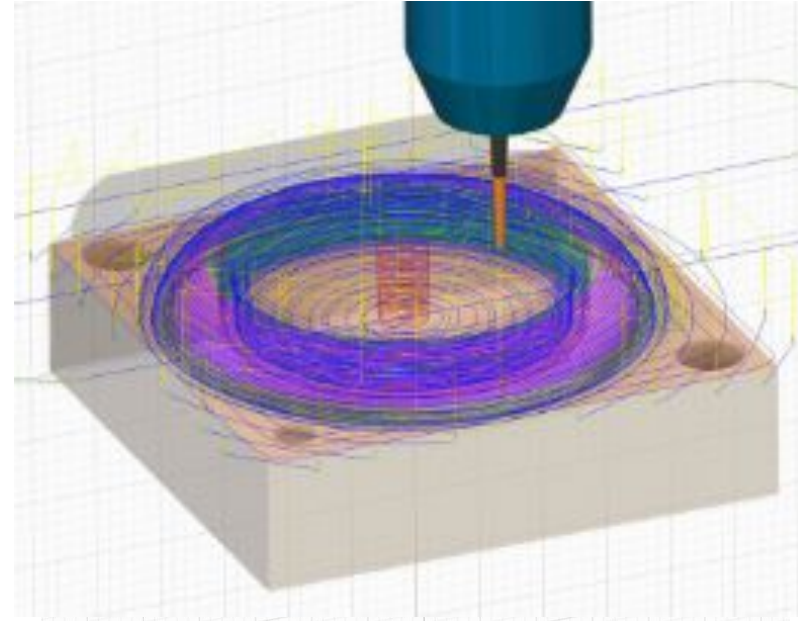
CAM Programs

Floatie - cavity

Time: 55 mins

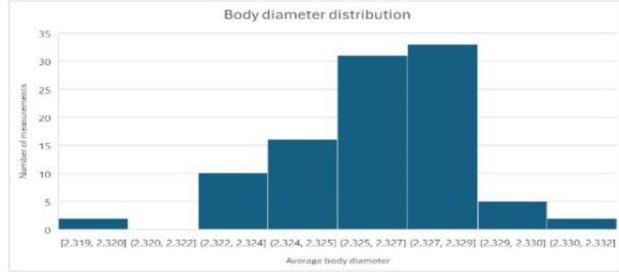


Operation	Instance	Setup	Work Offset	Tool
Face1	1/1	cavity	3	#14 - Ø2" 45° face
Adaptive4	1/1	cavity	3	#6 - Ø3/8" flat (End Mill)
Flat1	1/1	cavity	3	#6 - Ø3/8" flat (End Mill)
Contour1	1/1	cavity	3	#9 - Ø1/4" ball (End Mill)
Bore1	1/1	cavity	3	#1 - Ø1/16" flat (End Mill)
Bore2	1/1	cavity	3	#1 - Ø1/16" flat (End Mill)
Bore3	1/1	cavity	3	#10 - Ø1/16" 2° tapered (2 Degree Tapered End Mill)



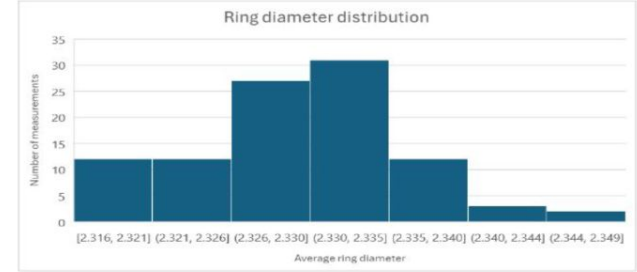
Kir-B

Press Fit Analysis

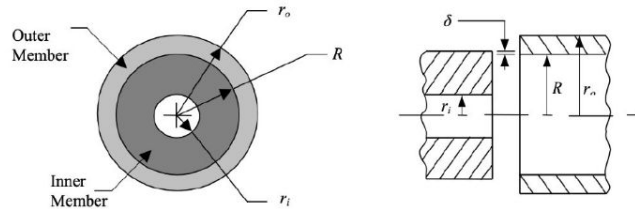


Empirical

	Interference
Team A	0.005131666
Team B	-0.0035
Team C	0.024495667
Team D	0.007320333
Team E	0.016



2.008 Yo-yo Projects - Spring 2024 Production Data



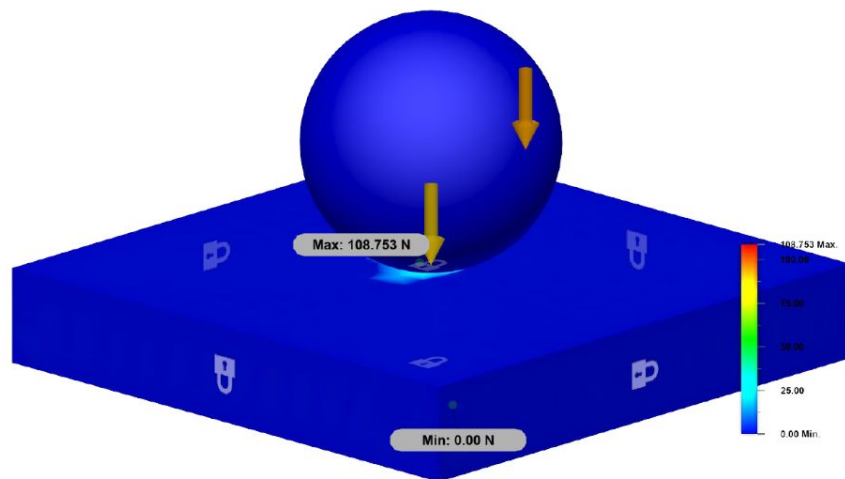
Theoretical

$$\delta_{\text{total}} = [K_o + K_i] p R$$

$$K_i = \frac{1}{E_i} \left[\left(\frac{R^2 + r_i^2}{R^2 - r_i^2} \right) - \nu_i \right] = \frac{1}{E_i} [C_i - \nu_i] \quad C_i = \frac{\xi_i^2 + 1}{\xi_i^2 - 1} \quad \xi_i = R / r_i$$

$$K_o = \frac{1}{E_o} \left[\left(\frac{r_o^2 + R^2}{r_o^2 - R^2} \right) + \nu_o \right] = \frac{1}{E_o} [C_o + \nu_o] \quad C_o = \frac{\xi_o^2 + 1}{\xi_o^2 - 1} \quad \xi_o = r_o / R$$

Press Fit Analysis



Typical Engineering Properties of Polypropylene

General Properties	English Units	SI Units
CAS Number	9003-07-0	9003-07-0
Density		
Homopolymer	56.4 - 56.6 Lbs /ft ³	0.904 - 0.908 g/cm ³
Random Copolymer	56.4 - 56.6 Lbs /ft ³	0.904 - 0.908 g/cm ³
Impact Copolymer	56.0 - 56.2 Lbs /ft ³	0.898 - 0.900 g/cm ³
TPOs	54.6 - 56.1 Lbs /ft ³	0.875 - 0.880 g/cm ³
Melt Density	9.5 lbs/in ³	0.739 g/cm ³
Bulk Density		
Pellets	32 - 36 lbs./ft ³	513 - 577 kg/m ³
Flake	29 - 31 lbs./ft ³	465 - 497 kg/m ³
Permeability Coefficients:		
Water (@25°C)	6.0x10 ⁻⁹ in ² /sec ² -atm	5.1x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Oxygen (@ 30°C)	2.7x10 ⁻⁹ in ² /sec ² -atm	2.3x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Carbon Dioxide (@30°C)	10.8x10 ⁻⁹ in ² /sec ² -atm	3.5x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Hydrogen (@ 20°C)	48.3x10 ⁻⁹ in ² /sec ² -atm	41.0x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Nitrogen (@30°C)	0.52x10 ⁻⁹ in ² /sec ² -atm	0.27x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Water Absorption @24 h Immersion	<0.01%	<0.01%
Brittleness Temperature		
Homopolymer	13 - 32 °F	-10 - 0 °C
Random Copolymer	-5 - 20 °F	-20 - -5 °C
Impact Copolymer	-44 - 20 °F	-40 - -5 °C
Mechanical Properties		
Modulus of Elasticity (Young's Modulus)		
Homopolymer	183,000 psi	1,300 MPa
Copolymer	155,000 psi	1,100 MPa
Poisson's Ratio	0.42	0.42
Hardness		
Shore D Scale	55 - 65	55 - 65
Rockwell R Scale		
Homopolymers	80 - 110	80 - 110
Random and Impact Copolymers	45 - 95	45 - 95
Coefficient of Friction, Plastics to Steel		
Static	0.30	0.30
Dynamic	0.28	0.28
Coefficient of Friction, Plastics to Plastic		
Static	0.76	0.76
Dynamic	0.44	0.44
Thermal Properties		
DSC Melting Point		
Homopolymer	320 - 329 °F	160 - 165 °C
Copolymer	275 - 318 °F	135 - 159 °C
Specific Heat (@ 23°C)		70 -80 J°K/mol
Heat of Fusion	37.8 Btu/lb	88 kJ/kg
Thermal Conductivity (solid)		0.17 - 0.22 W/m °K

Press Fit Analysis

Basic Params for Calculations

```
SF = 10; % safety factor [-]
in_to_m_conversion = 39.37;
E = 1300 * 10^6; % young's modulus [Pa]
nu = 0.42; % poisson ratio [-]
```

Energy Calcs for Contact Force Simulation in Fusion

```
rough_mass = .03; % kg
g = 9.81; % m/s^2
h = 47.5 / in_to_m_conversion; % inches above floor for drop test simulation

init_vel = sqrt(2*g*h); % (m/s) initial velocity for fusion simulation

contact_force = 100.8; % N of force at sphere-floor interface
mu_s_p2p = .76; % Static plastic-to-plastic coefficient of friction for PP
```

Shigley's Interference Calculations {02/27/25}

```
R = 1.25 / in_to_m_conversion; % nominal radius [m]
r_o = 1.5 / in_to_m_conversion; % outer radius [m]
r_i = 1.24 / in_to_m_conversion; % inner radius [m]
length_of_engagement = 0.2 / in_to_m_conversion;
```

```
zeta_o = r_o/R;
zeta_i = r_i/R;
```

```
C_o = (zeta_o^2 + 1)/(zeta_o^2 - 1);
C_i = (zeta_i^2 + 1)/(zeta_i^2 - 1);
```

```
K_o = (1/E) * (C_o + nu); % constant for yo-yo ring
K_i = (1/E) * (C_i + nu); % constant for yo-yo body
```

```
A_wall = 2*pi*R*length_of_engagement;
```

```
p_sim = contact_force/(A_wall * mu_s_p2p); % contact pressure at wall
p_yield = 33*10^6; % contact pressure at wall
```

```
interference_yield = SF * ((K_o + K_i) * p_yield * R)/in_to_m_conversion
interference_yield =
-0.0242
```

```
interference_sim = SF * ((K_o + K_i) * p_sim * R)/in_to_m_conversion
```

```
interference_sim =
-1.0351e-04
```

Name	Unit	Expression	Value	Comments
wall_thickness	in	0.08 in	0.08	uniform wall thickness
outer_radius	in	1.25 in	1.25	outer radius of kirby
inner_radius	in	outer_radius - wall_thickness	1.17	inner radius of kirby
length_of_engagement	in	0.2 in	0.2	press fit engagement
total_outer_radius	in	outer_radius + floatie_radius	1.75	total size of floatie
nut_spacer_width	in	0.25 in	0.25	space for nut spacer
nut_spacer_height	in	(1 / 8) * 1 in	0.125	space for the nut spacer
floatie_radius	in	0.50 in	0.5	radius of just the floatie
interference	in	0.0242 in	0.024	interference for press fit
foot_heel_radius	in	0.45 in	0.45	radius of the circle composing kirby's foot
foot_length	in	1.65 in	1.65	length of kirby's foot from heel to toe
shrinkage_scaling		(1.0152 in) / in	1.0152	PP shrinkage
draft_angle	deg	2 deg	2	draft angle
floatie_fillte_radius	in	0.13 in	0.13	top edge fillet



interference

in

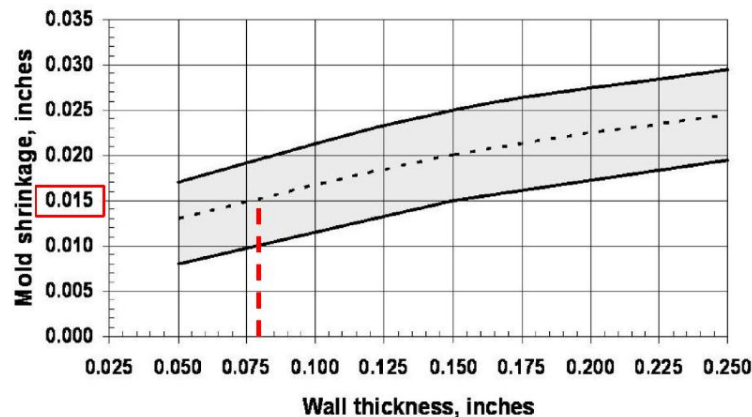
0.0242 in

0.024 interference for press fit

Kir-B

Shrinkage Analysis

Figure 1: Shrinkage of polypropylene



$$\text{scaling factor} = \frac{1}{1 - \% \text{ shrinkage}}$$

shrinkage_scaling

(1.0152 in) / in

1.0152 PP shrinkage

All parts use shrinkage_scaling in CAD

Name	Unit	Expression	Value	Comments
wall_thickness	in	0.08 in	0.08	uniform wall thickness
outer_radius	in	1.25 in	1.25	outer radius of kirby
inner_radius	in	outer_radius - wall_thickness	1.17	inner radius of kirby
length_of_engagement	in	0.2 in	0.2	press fit engagement
total_outer_radius	in	outer_radius + floatie_radius	1.75	total size of floatie
nut_spacer_width	in	0.25 in	0.25	space for nut spacer
nut_spacer_height	in	(1 / 8) * 1 in	0.125	space for the nut spacer
floatie_radius	in	0.50 in	0.5	radius of just the floatie
interference	in	0.0242 in	0.024	interference for press fit
foot_heel_radius	in	0.45 in	0.45	radius of the circle composing kirby's foot
foot_length	in	1.65 in	1.65	length of kirby's foot from heel to toe
shrinkage_scaling		(1.0152 in) / in	1.0152	PP shrinkage
draft_angle	deg	2 deg	2	draft angle
floatie_fillte_radius	in	0.13 in	0.13	top edge fillet



Press Fit Specifications

Specifications

- * variability introduced through:
 - Different people gathering measurements
 - Taking measurements while pieces warm
- * engineering reasoning
 - Machining tolerances and discrepancies
 - Shrinkage

Floatie

Body Average:	2.504596667	[in]
Body Std Dev:	0.009050679731	[in]

Butt

Ring Average:	2.512073333	[in]
Ring Std Dev:	0.008684778791	[in]

Head

Ring Average:	2.507661749	[in]
Ring Std Dev:	0.008208960144	[in]

Success!

- The pieces fit together!
- 9/10 passed the drop test